AIRCRAFT MAINTENANCE MANUAL (AMM) - BOEING 737 SERIES
DOCUMENT REF: AMM-B737-REV42
SECTION 1: DAILY INSPECTIONS & PRE-FLIGHT MAINTENANCE PROCEDURES

1.1 GENERAL MAINTENANCE POLICY
All aircraft operating under institutional airworthiness regulations must undergo scheduled and unscheduled maintenance according to the approved Maintenance Control Program. Maintenance records must be preserved for minimum 24 months.

1.2 PRE-FLIGHT WALK-AROUND INSPECTION
Before each departure, certified maintenance personnel or the flight crew must perform an external walk-around inspection adhering to standard checklist sequence:
- Pitot-static probes: Check covers removed, openings unobstructed by foreign objects or ice.
- Radome and forward fuselage: Inspect for bird strike impacts, hail damage, or skin wrinkles.
- Landing gear assembly:
  * Check main landing gear shock strut extension (nominal 4.5 to 5.5 inches).
  * Inspect tire condition: Minimum tread groove depth is 2.0 mm; verify absence of deep cuts.
  * Check brake wear indicator pins; replace brake assembly if pin is flush with housing.
  * Main tire pressure must be maintained at 205 ± 5 PSI at cold ambient temperatures.
- Hydraulic systems: Verify Reservoirs A, B, and Standby are pressurized above 2,800 PSI.
- Engine cowlings and fan blades: Visually inspect all 24 titanium fan blades for nicks, cracks, or leading-edge deformation. Minimum allowable oil quantity before dispatch is 12 quarts.

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SECTION 2: SCHEDULED MAINTENANCE INTERVALS (A, B, C, AND D CHECKS)

2.1 MAINTENANCE INTERVAL CLASSIFICATIONS
To maintain airworthiness certification, maintenance is grouped into progressive check cycles:

- A-CHECK (Every 400 to 600 Flight Hours or 60 Days):
  Routine visual inspection performed in hangar overnight. Includes lubrication of flight control surfaces, inspection of emergency evacuation slide inflation cylinders, checking engine fuel filters, and testing APU (Auxiliary Power Unit) fire detection loops.

- B-CHECK (Every 6 to 8 Months):
  Detailed mechanical alignment check. Calibration of flight instruments, electrical bus isolation testing, and inspection of wheel well wiring harnesses.

- C-CHECK (Every 20 to 24 Months or 6,000 Flight Hours):
  Comprehensive structural and systems overhaul requiring aircraft grounding for 1 to 2 weeks. Involves non-destructive ultrasonic testing (NDT) of wing spar attach fittings, flap track fairing actuators, horizontal stabilizer jack screw lubrication, and complete cabin pressurization leak test.

- D-CHECK (Every 6 to 10 Years):
  Heavy maintenance visit (HMV). The aircraft is entirely stripped of paint, interior cabin removed, engines dismounted, and the fuselage undergo complete metallurgical fatigue testing.

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SECTION 3: HYDRAULIC ANOMALIES & MANUAL GEAR EXTENSION MAINTENANCE

3.1 HYDRAULIC SYSTEM REDUNDANCY
The aircraft operates with two independent primary hydraulic systems (System A and System B) and an auxiliary Standby System. Nominal operating pressure is 3,000 PSI (+/- 100 PSI).

3.2 MANUAL GEAR EXTENSION MECHANISM
In the event of total dual hydraulic pressure loss (System A and System B failure), the landing gear must be deployed via the manual extension handles located on the flight deck floor behind the first officer.
- Manual extension relies on gravitational free-fall and aerodynamic air load locking.
- Maintenance Inspection Protocol for Manual Extension Cables:
  1. Inspect stainless steel tension cables for frays, kinks, or corrosion every C-Check.
  2. Cable tension must be calibrated to 45 lbs (+/- 3 lbs) at 20 deg C using calibrated tensiometer.
  3. Uplock mechanical release cams must be lubricated using AeroShell Grease 22.
  4. Perform functional gear free-fall swing test on hangar jacks following any component replacement.